Test

If the geothermal gradient at Rosemanowes is 37°C/km, and the surface temperature averages 10°C, what is the minimum depth to which drilling must penetrate to obtain a temperature of 150°C (i.e. viable for geothermal power)?

Δ from surface to 150°C: 140°C

140 / 37 = **3.784 km**

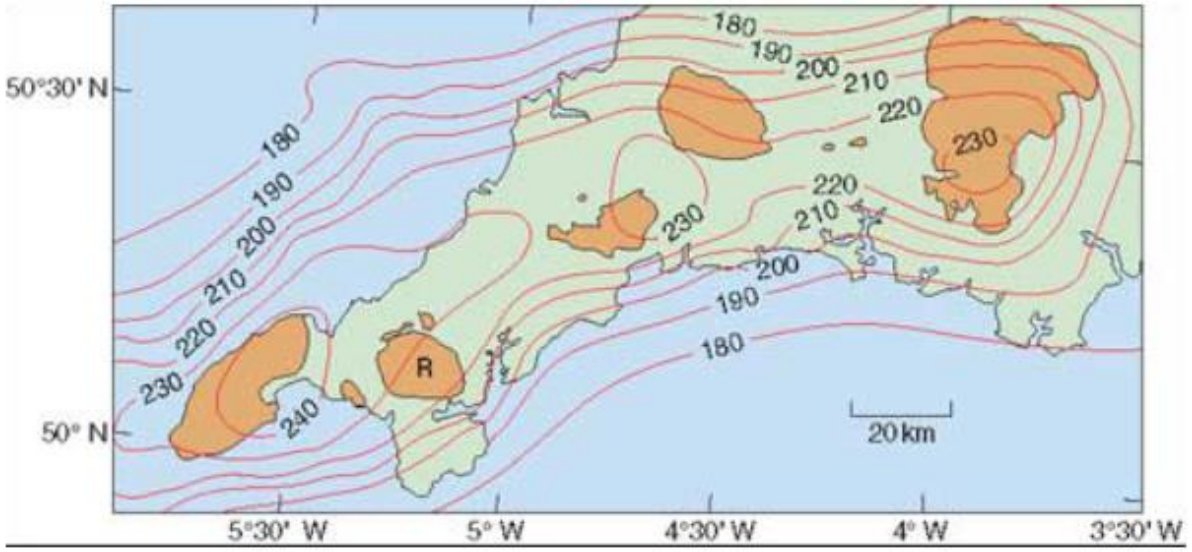


Figure 7 Surface occurrences of granite in Cornwall, which are connected at depth to form a much larger mass. Also shown are contours of the temperature in °C that are expected 6 km beneath the surface. The Rosemanowes HDR experiment is at R.

Test

Appraisal drilling for a newly discovered copper deposit has identified a horizontal ore body 500 m long, 200 m wide and 10 m thick. Density of the ore body: 2.7 g/cm^3 = t/m^3. Grades determined from drilling:

Grades (%)
2.15
1.52
1.45
2.07
1.05
1.26
1.77
2.69
2.60
1.20
3.05
3.58
1.70
2.38
1.53

Determine the mass of copper than can be recovered, as well as its market value for low-average-high cases. Copper market price is $9.5/kg, production cost $6/kg.

Volume of ore body: 500 m x 200 m x 10 m = 1,000,000 m^3

Tonnage = 1,000,000 m^3 x 2.7 t/m^3 = 2,700,000 t of rock

Average grade?
2%

What about low/high cases?
Standard deviation
SD = 0.74 => high case 2.74%, low case 1.26%

Average case: Mass of Cu mined: 2,700,000 t x 0.02 = **54,000 t Cu**
Low case: Mass of Cu mined: 2,700,000 t x 0.0126 = **34,020 t Cu**
High case: Mass of Cu mined: 2,700,000 t x 0.0274 = **73,980 t Cu**

$9.5/kg market price, OPEX cost to produce is $6/kg
Average case: 54,000,000 kg x (9.5-6) = **189,000,000 $**
Low case: 34,020,000 kg x (9.5-6) = **119,070,000 $**
High case: 73,980,000 kg x (9.5-6) = **258,930,000 $**

Test

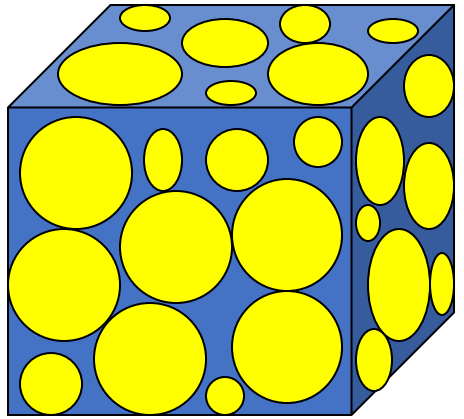
An isotope of technicium, technicium-99, has a half-life of 6 hours. A hospital has a sample which contains 40 mg of technicium-99. Calculate how much technicium-99 will be in this sample after 12 hours.

After 1 half life (6 hours) the amount of technicium-99 will have halved from 40 mg to 20 mg. After another 6 hours, it will have halved again to 10 mg.

Test

Calculate the grain volume (in cm^3) and porosity of a cylindrical sand pack. Available data is given as follows: length = 50 cm; diameter = 6 cm; weight of dry sand = 2770 g; and the grain density of the dry sand = 2.6 g/cm^3.





$Porosity = V_{pore} / V_{bulk}$

$V_{bulk} = \pi \times r^2 \times L\ (cm^3)$

$= 1414.1\ cm^3$

$V_{pore} = ?$

$V_{pore} = V_{bulk} - V_{grain}$

$V_{grain} = mass\ (g) / \delta_{grain}\ (g/cm^3)$

$= 2770\ g / 2.6\ g/cm^3 = 1065.4\ cm^3$

$Porosity = V_{pore} / V_{bulk} = (V_{bulk} - V_{grain}) / V_{bulk} = (1414.1 - 1065.4) / 1414.1 = 348.7 / 1414.1 = 0.247$

Exam

Name 2-3 evidences why diamonds derive from the mantle.

Density, crystal structure, mantle inclusions, found in rocks that come from mantle

Exam

Which factor(s) determine whether an element becomes concentrated in the solid phase during melt crystallization?

A. Its density
B. Its conductivity
C. Its preference for silicate, iron, sulfur bonds
D. its susceptibility to magnetic fields
E. its compatibility with crystallizing minerals

Correct: C, E

Exam

Discuss **two** processes causing metal precipitation from hydrothermal fluids.

Any from:
wall rock alteration,
boiling,
mixing with meteoric water,
temperature drop,
pressure drop,
changes of redox state

Exam

What are the key geological factors required for a conventional petroleum accumulation?
Why is it important that all are present and develop in the correct order?

Source rocks, generation/migration, reservoir, seal and trap

If one not present at the correct time, the play will not work.

Exam

When is drilling usually employed in the life of a commodity exploration programme? Briefly outline 3 reasons for drilling.

In the later phases of an exploration program, after a deposit is discovered (or at least suspected)

reasons:
deposit identification,
assess size of deposit,
assess quality/grade of deposit,
obtain samples for further study

Exam

Exploration mapping in a basin in Brazil has delivered the following results:

The northern half of the area exposes a thick succession of shales that pass laterally into limestones with corals. This succession is dipping southwards into the subsurface.

The southern part of the area does not contain any outcrops, but it contains several subsurface mines (by different companies) that exploit potash and rocksalt from evaporites present in the subsurface.

One of your geologists is a keen amateur speleologist and was delighted to find several large caves in the limestones during mapping. However, he was annoyed that it cost him a pair of field pants – he slipped in a puddle of tar that had seeped up and collected on the floor of one of the caves.

Questions:

1. In light of the results of the mapping and field observations, what type of new deposit was the company exploring for? Where should the company direct its exploration in the future? Explain your answers.
2. Develop a genetic model for the deposit type targeted by the company. Support this model with evidence gathered in the field and explain what type of information you are hoping will further support your model from a suite of samples that you have sent to the University of Manchester for study and analysis.

Oil accumulation in limestones, from what are likely black shales/source rocks in basin, generated oil during subsidence, migrated into what could be traps in porous shallow-water carbonates and below evaporite seals
Need to investigate source rocks, analysis of TOC, maturity; need to investigate limestones (porosity).